Project 2026
BINF-F401

Vincent Detours

Overview of course evaluation

Written exam

- Questions assess your understanding of the theory
- It’s an open book exam or 2 hours (but don’t expect to study the course during these two hours!)
- You’ve got a personal grade

Project

- You apply ideas and tools of the course on real data
- It’s a mini research project evaluated from a written report
- It’s done by groups of two students, so grade is collective

Final grade is the average of exam and project, but you need to have at least 8/20 in each part of the evaluation.

Biological aging clocks

Aging is a controlled process more than a passive decay

- Metabolic/genetic manipulations increase C. Elegans life span by 7-fold (!!!).

- Caloric restriction increases life span in several mammal species

- Rapamycin fed late in life increased the life span of mice (if it worked in human we could live 120yrs, starting rapamycin at 40)

Does aging protect against cancer ?

- Our DNA accumulate mutations as we age and mutations may cause cancer

- Several progeria syndromes are caused by defects in DNA repair genes

- These may affect metabolism via the IGF1 pathway

- Perhaps ageing is a process activated by DNA damage to protect us against cancer?

Chronological vs. biological age

- Some people look younger (or older) than others with the same chronological age
- E.g. a life under a dry sunny climate makes the skin look wrinkled, i.e. older
- What are the determinant of *biological* ageing? Are they partly inherited?
- Is biological age a predictor of life span?
- Does biological age explain the onset of puberty and menopause?
- Are all our organs ageing at the same speed? Is organ age a predictor of organ failure?
- **To answering all these very basic questions we need to measure how fast the biological clock is ticking…**

The DNA methylation clock

- Steve Horvath (Genome Biology 2013, 14:R115; see also corrections published for this article) claim to have discovered a DNA methylation clock
- It accurately predicts the chronological age of normal individuals (median error <3.6yrs in validation dataset)
- It is applicable to methylation profiles of many organs
- (S. Horvath is a bioinformatician who worked only on published data!)

The DNA methylation clock

Some exciting preliminary findings

- DNAm age is accelerated during development, linear afterward
- Stem cells have a low DNAm age
- Cell passage increases DNAm age
- DNAm age *not* accelerated in progeria
- DNAm age accelerated in Down's sydrome
- DNAm age acceleration is highly inherited, especially during youth
- DNAm age could show some inter-organ variations (very limited data so far)

Telomere length

- Telomeres are chromosomes ends
- They are shortened at each cell division
- Their length is reset in germ cells
- Thus, telomere are biological cell division counters
- Hayflick limit: human cell can’t undergo >50 divisions

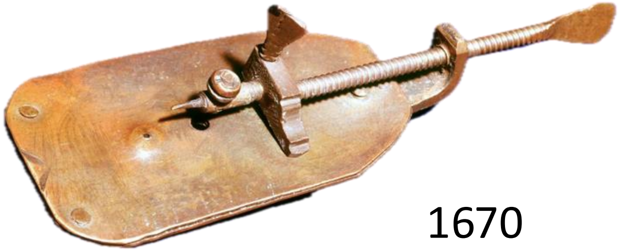
A computational framework for quantitative histology

Modern biology is born with the microscope

Van Leeuwenhoek, circa 1673
(replica)

The microscope is the core instrument of **histopathology**, the branch of medicine concerned with disease diagnostic from cells and tissues observation

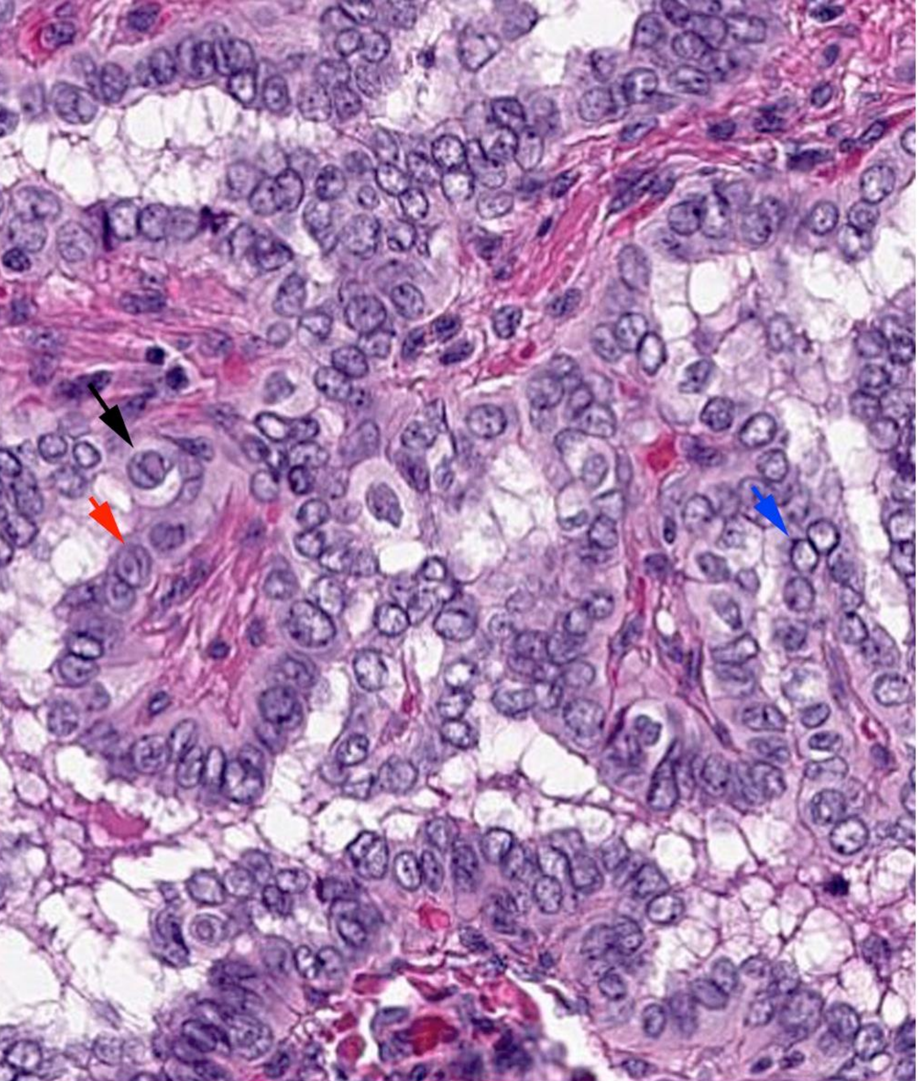
A fundamental limit of histopathology



1670

2026

- Microscope technology has evolved dramatically over the last 350 years
- By contrast, **histopathology descriptions have barely evolved: they remain qualitative and subjective**



Histopathology is qualitative and subjective

Example:

- Nuclei of thyroid cancer cells are said to be large, irregular, have the aspect of ground glass. Some look like coffee beans.

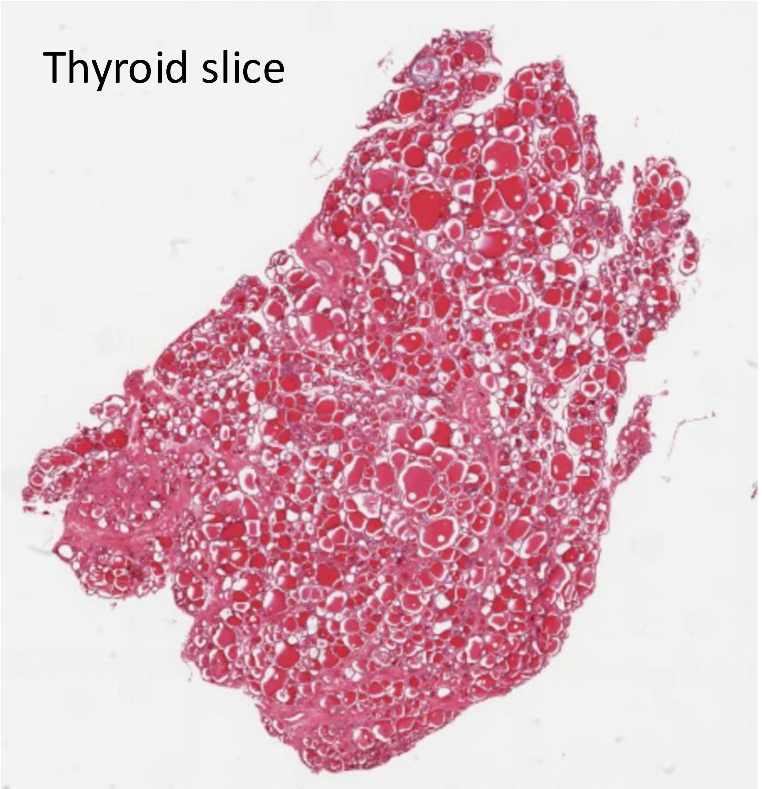
This verbal description is

- **Qualitative**: what is the limit between normal and 'big' nuclei?
- **Subjective**: do two pathologists have the same idea of 'big', 'irregular' and 'coffee bean-like'?

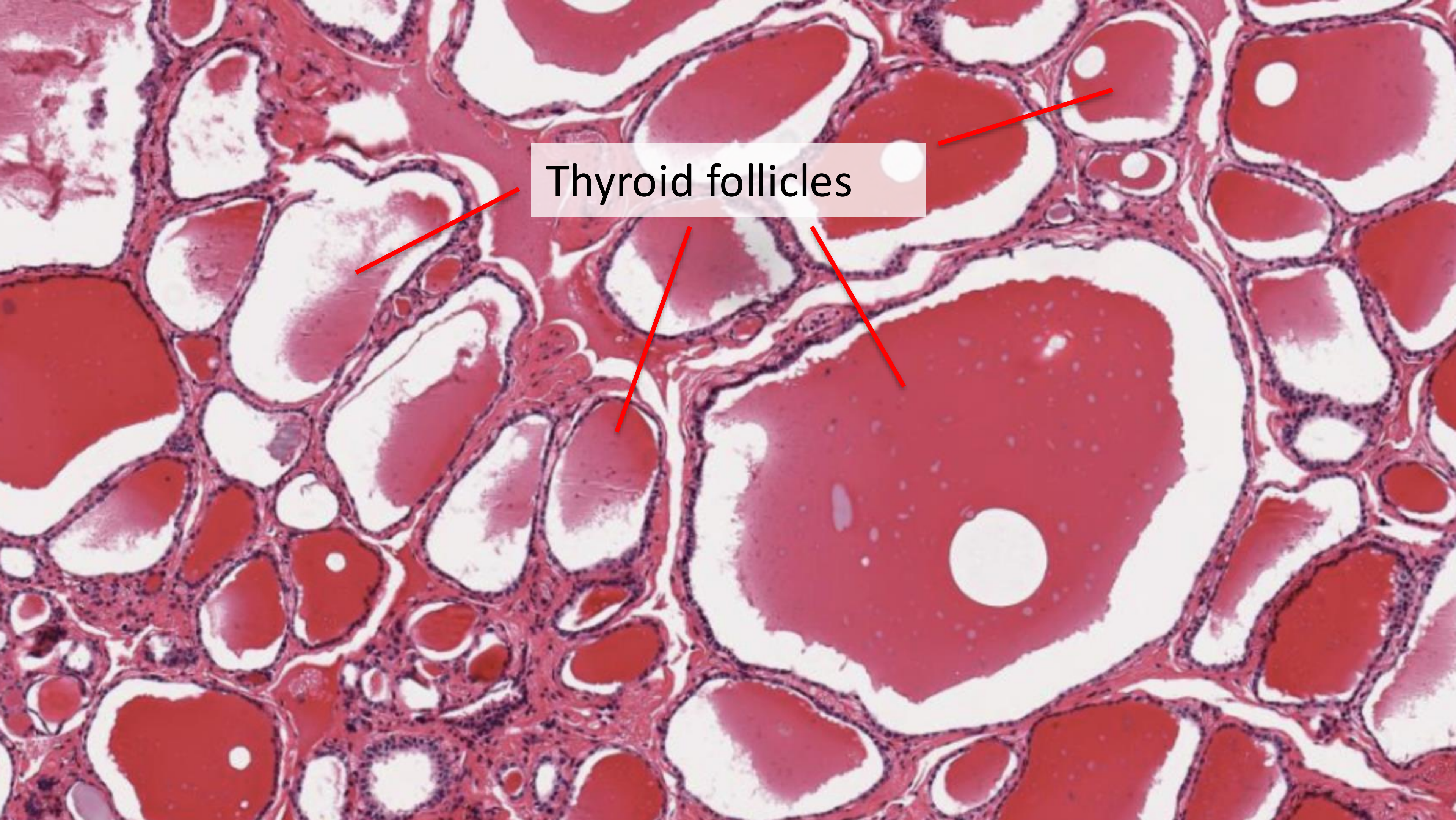
These shortcomings are obstacles to

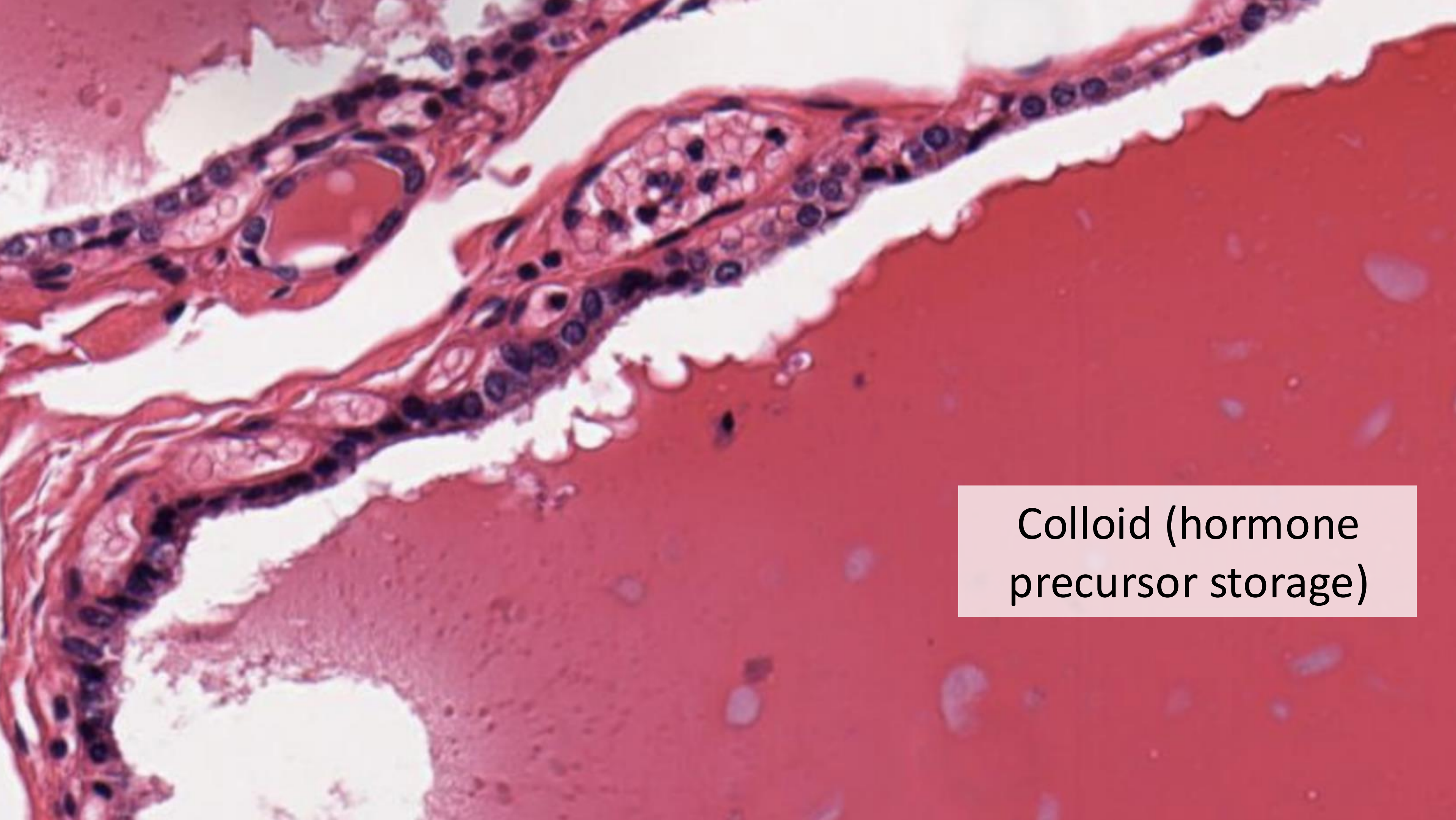
- the quality of diagnostics
- the scientific understanding of morphology

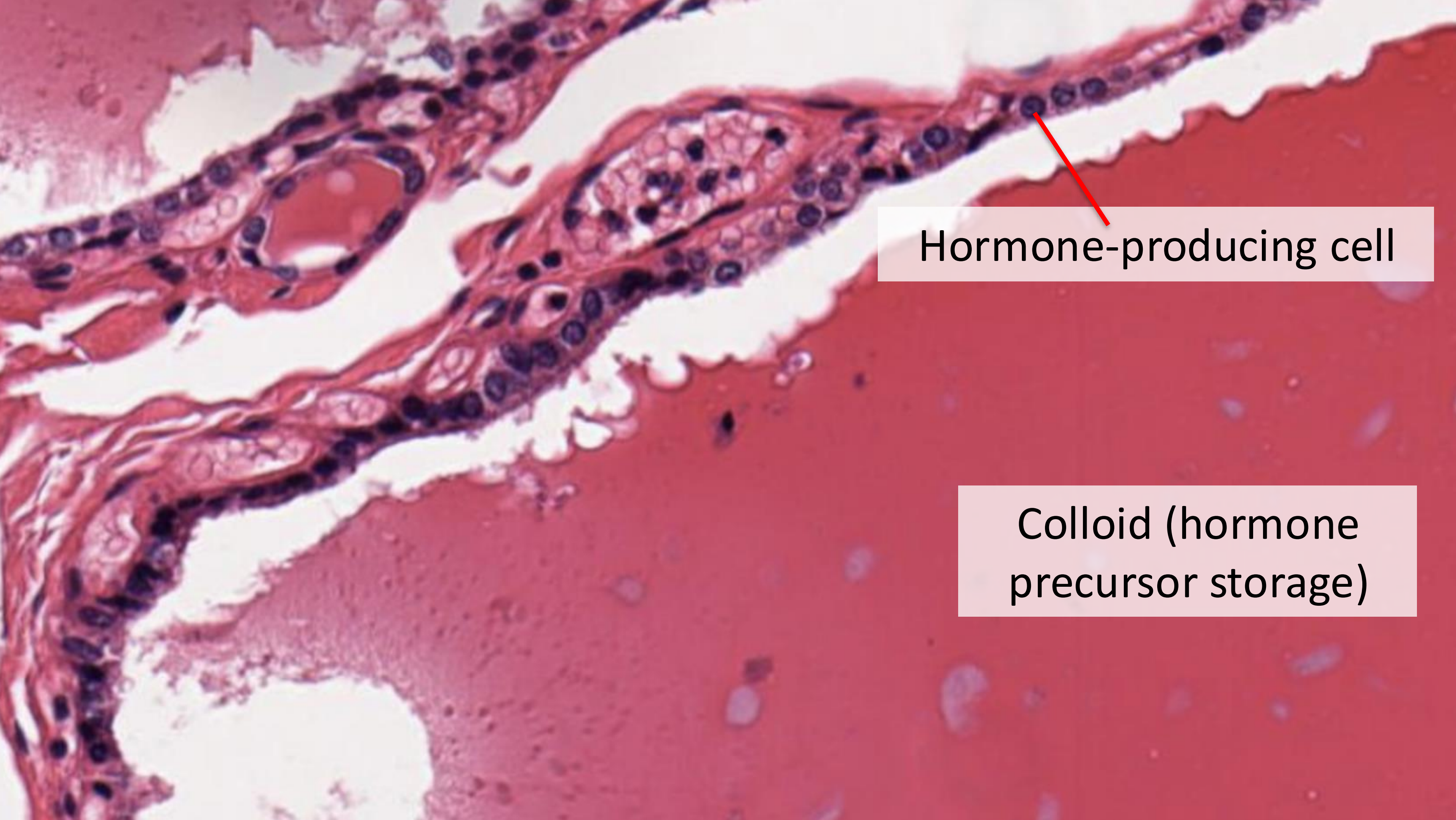
The raw material: high resolution whole slide images

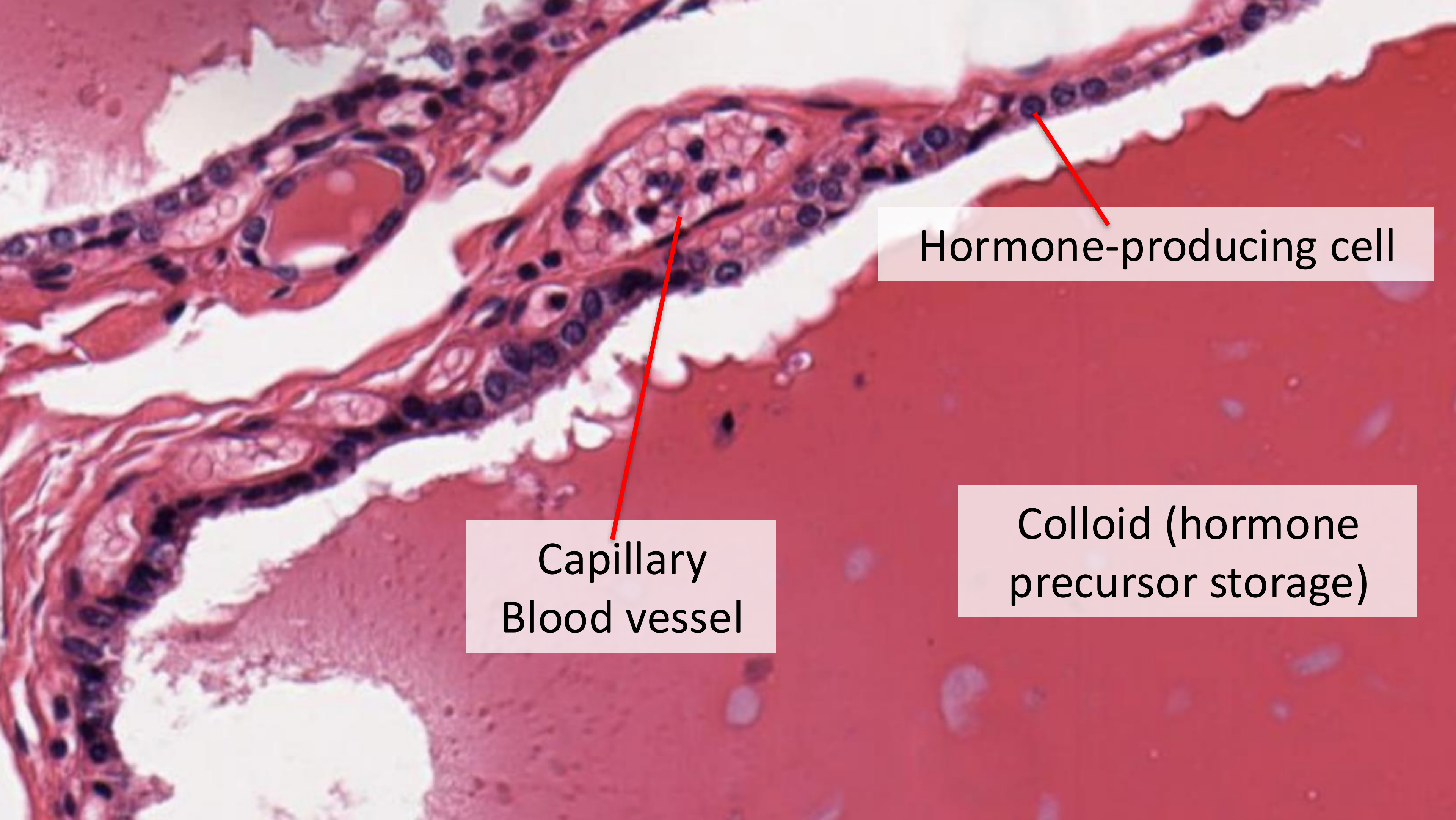


- Surgeons remove (part of) organ
- Histopathologists cut *slices*, 4-10 micron thick
- Slices are scanned at *very high resolution*

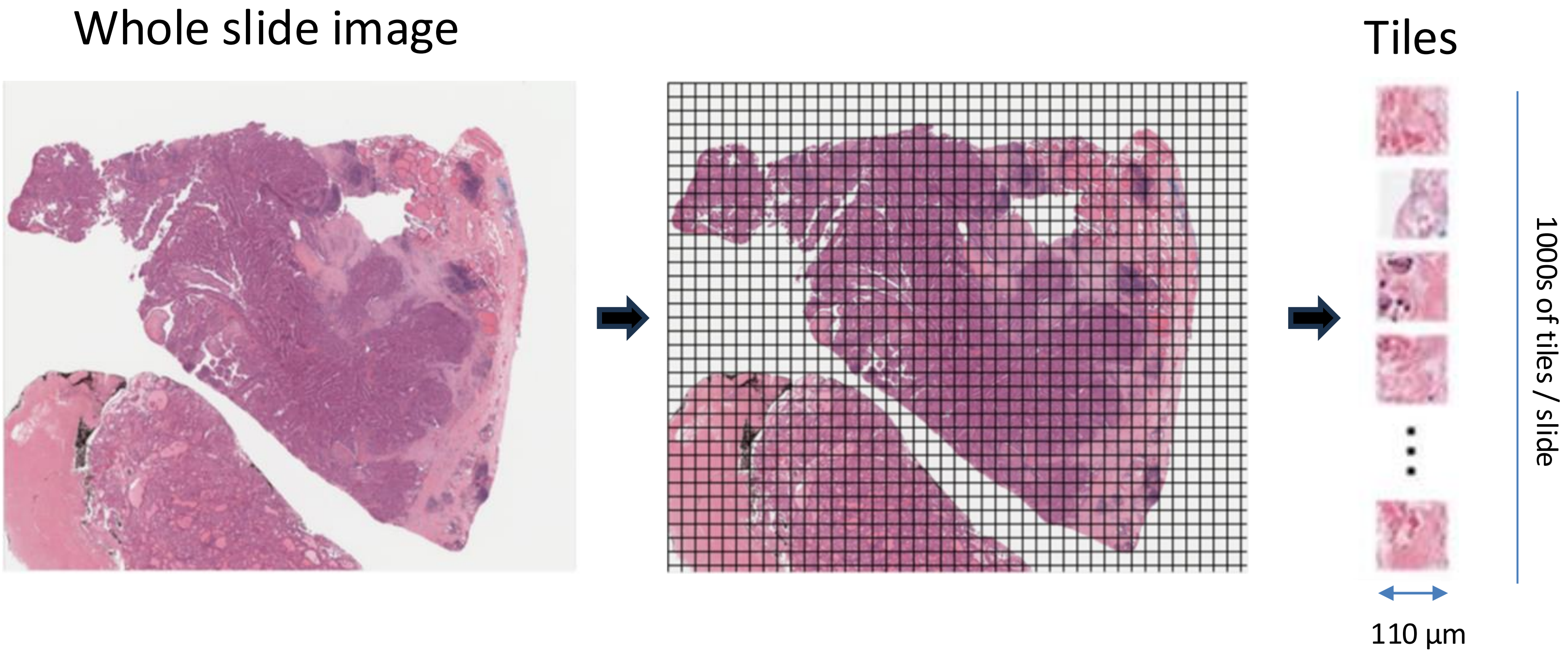

Thyroid follicles


Colloid (hormone
precursor storage)


Hormone-producing cell
Colloid (hormone precursor storage)

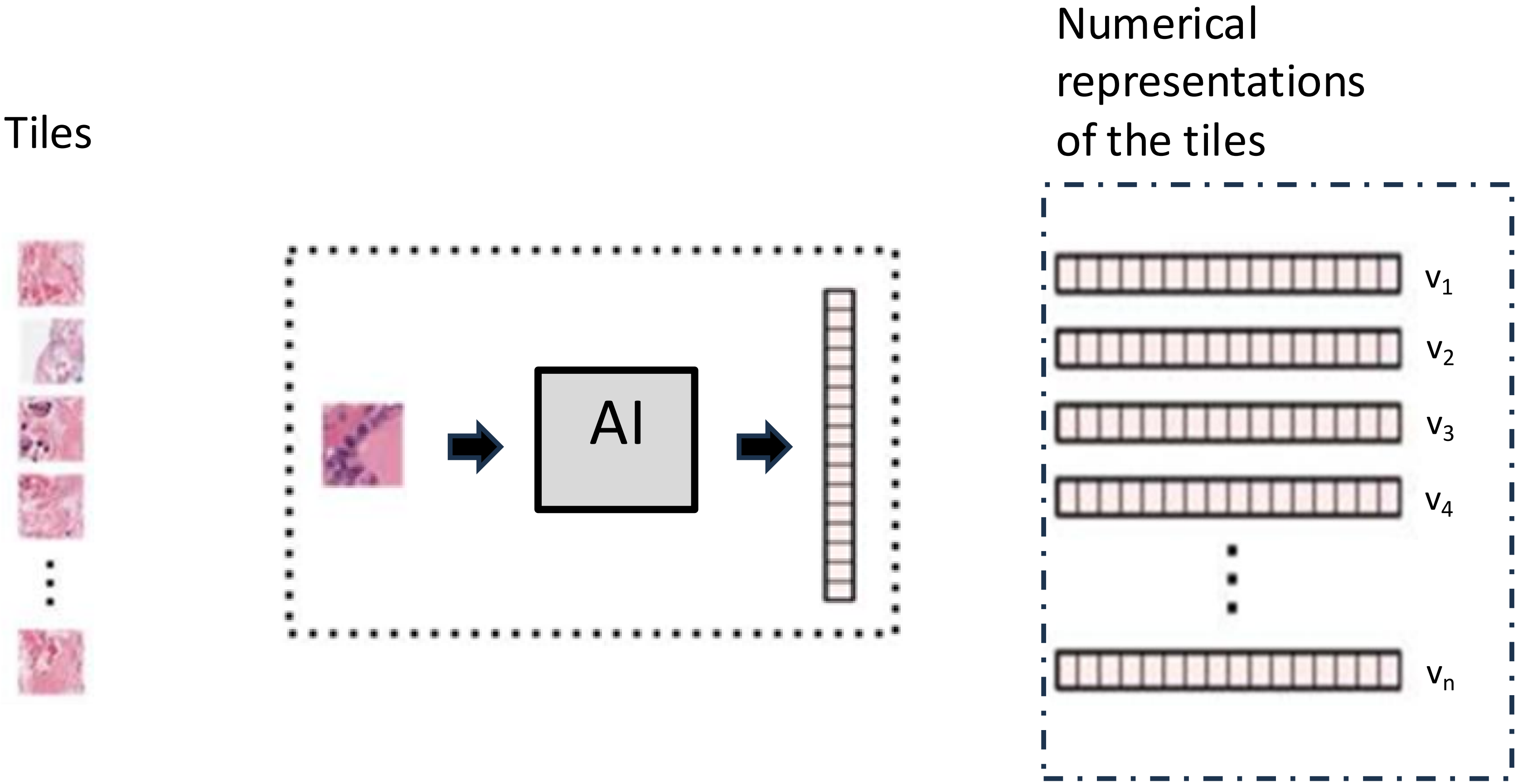

Hormone-producing cell
Capillary
Blood vessel
Colloid (hormone
precursor storage)

Step #1: tiling whole slide images

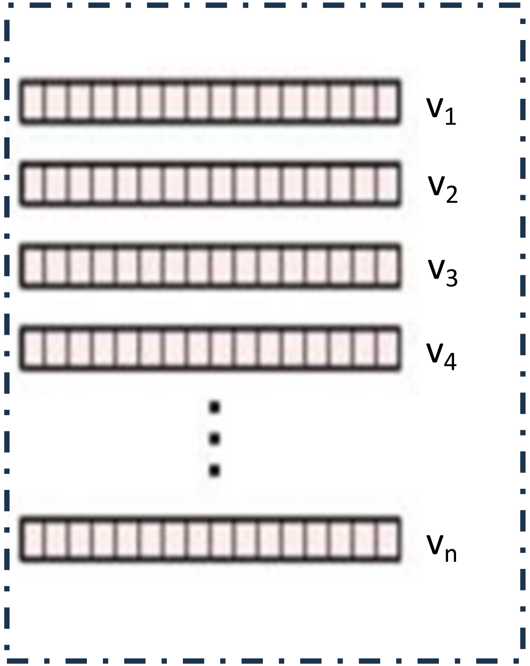


Whole slide images are tiled into 1000s of tiles of 110x110 μm^2

Step #2: compute numerical representations of tiles' content with an artificial intelligence



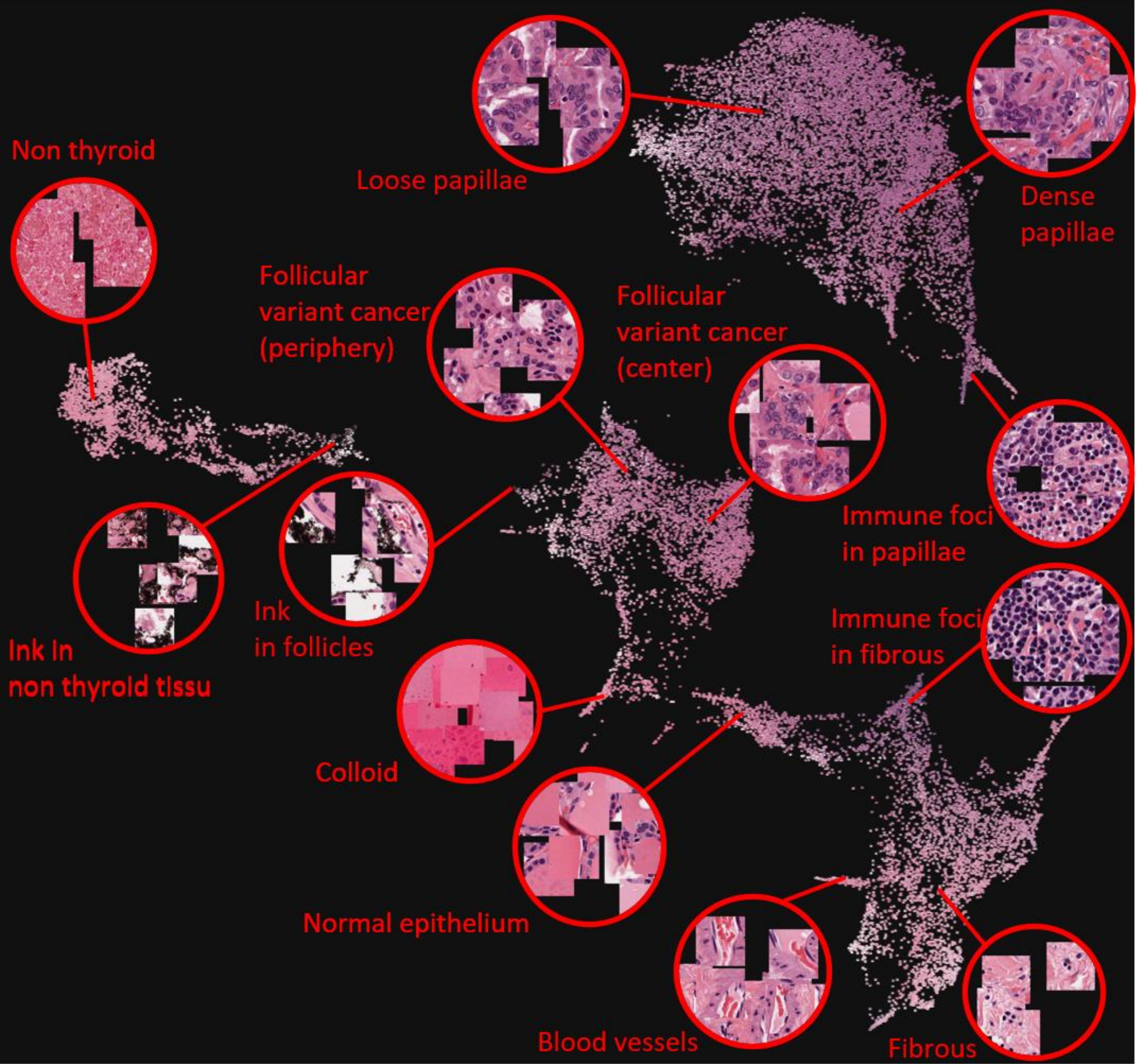
Step #2: compute numerical representations of tiles' content with an artificial intelligence



AI-generated vectors capture the *histopathological content* of the images

Because they are *numerical vectors* we can run *statistical computation* on them, including

- Dimensionality reduction (for visualisation)
- Clustering (to discover histopathological categories)



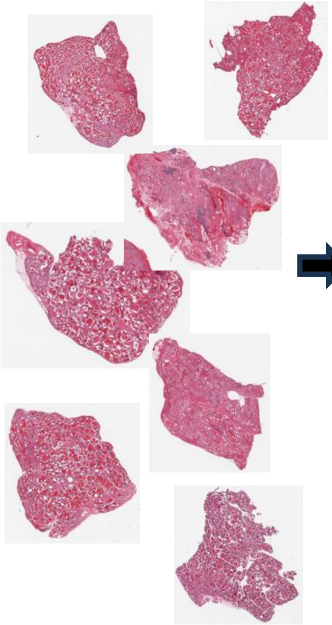
The AI places the tiles in a histological space

- Tile vectors define a mathematical space, that we call 'histological space'
- In this space, histologically similar tiles are close to one another
- One can identify **histological categories** as clusters of densely packed tiles

⬅ Tiles of PTC slide in (UMAP-reduced) histological space

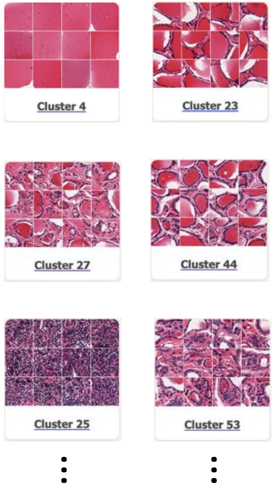
Step #3: identify population-scale histological categories

Large slide collection



Tiling
+
AI
+
Clustering

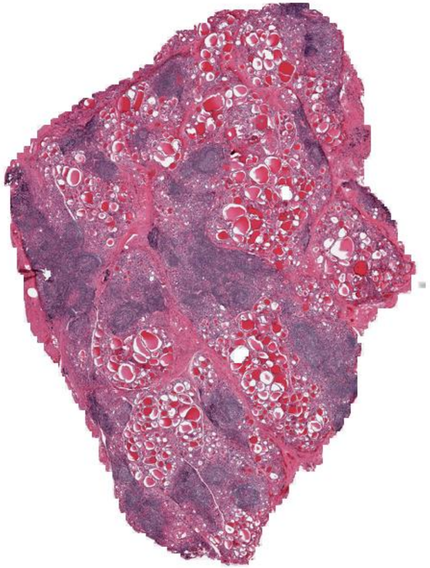
Histological categories



- The pipeline I just demonstrated on one slide is applied to *hundreds of slides from a population* of individuals
- The resulting categories define the *histological atlas* of all histological categories in that population

Step #4: we count tiles from each histological category in each individual slides

Slide

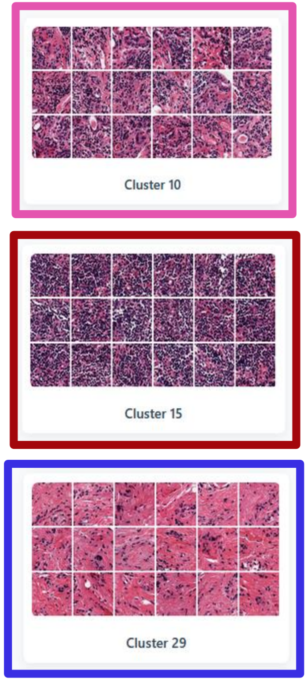


Step #4: we count tiles from each histological category in each individual slides

Slide

+

Histological categories



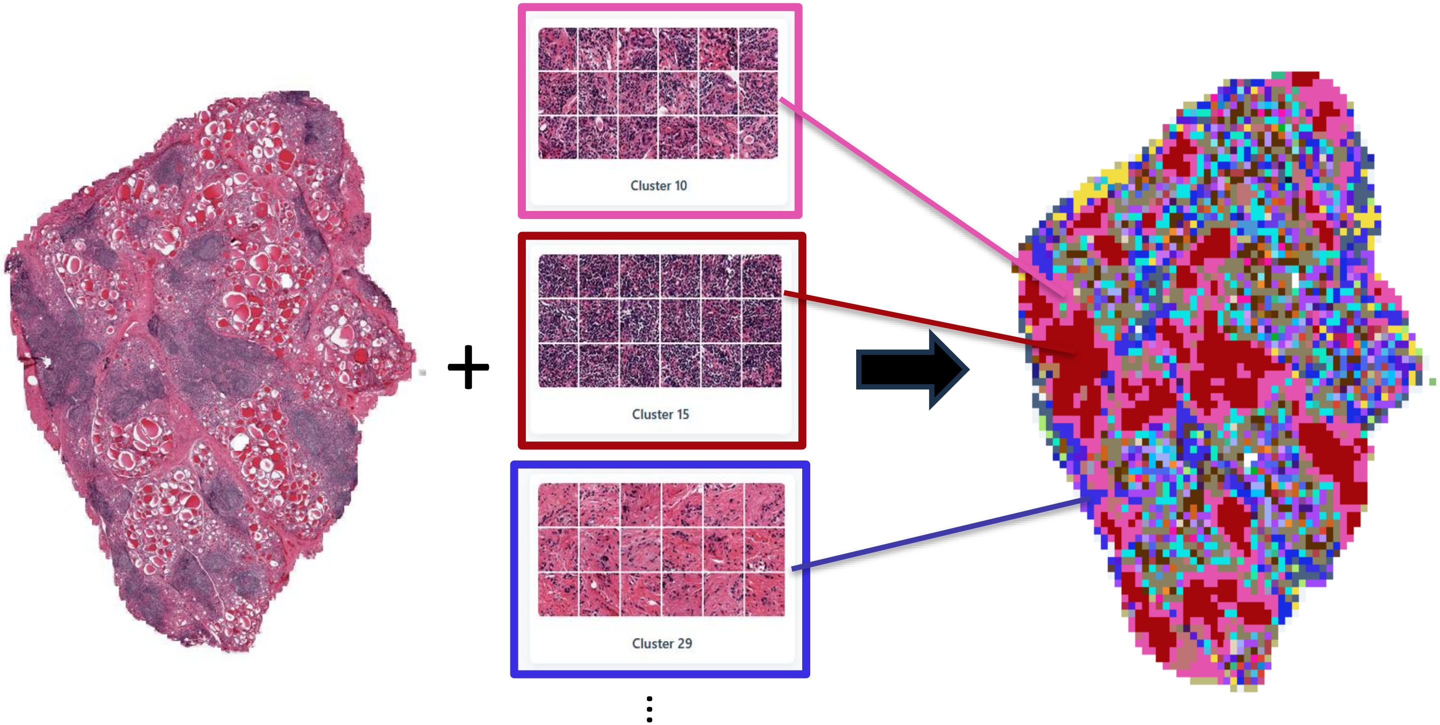
⋮

Step #4: we count tiles from each histological category in each individual slides

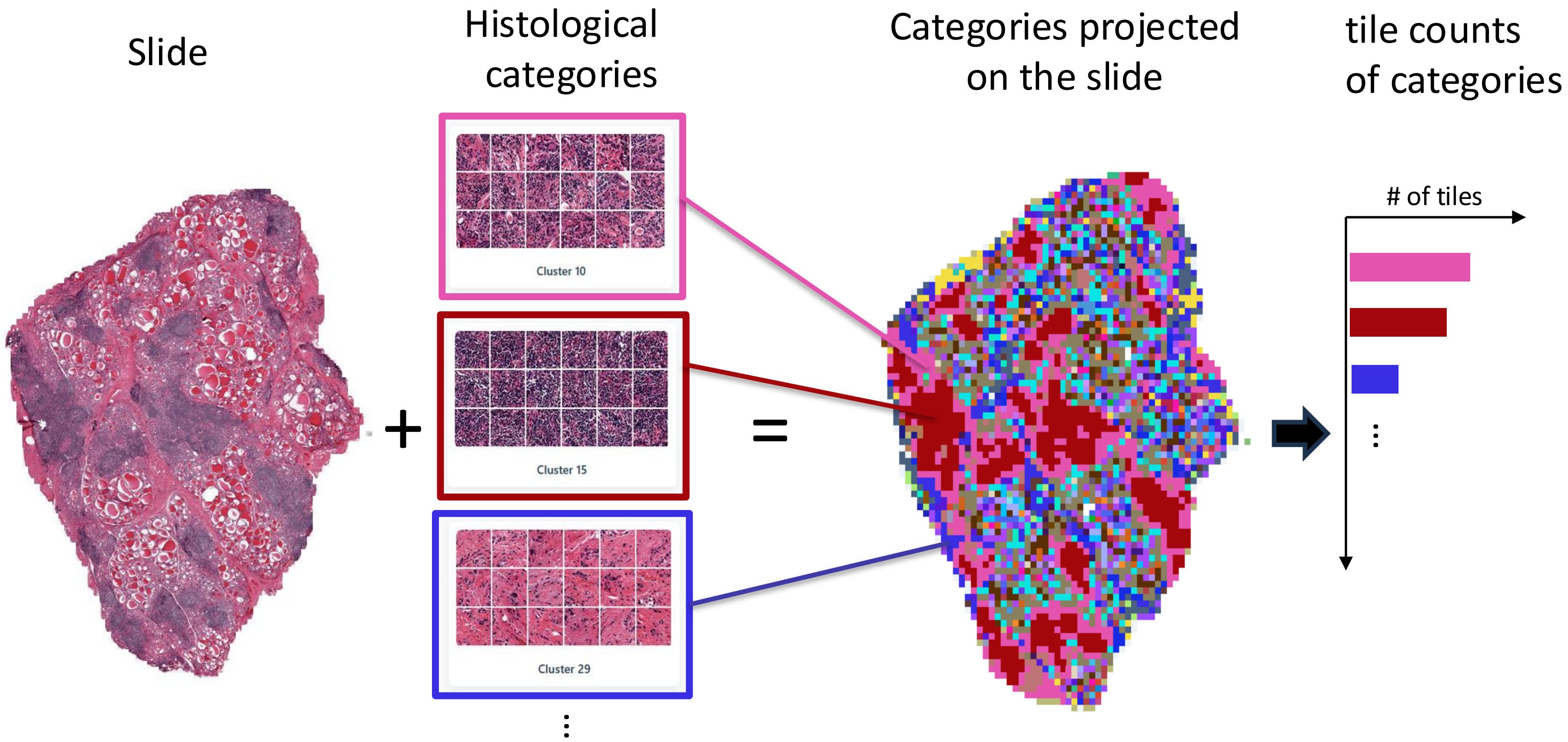
Slide

Histological categories

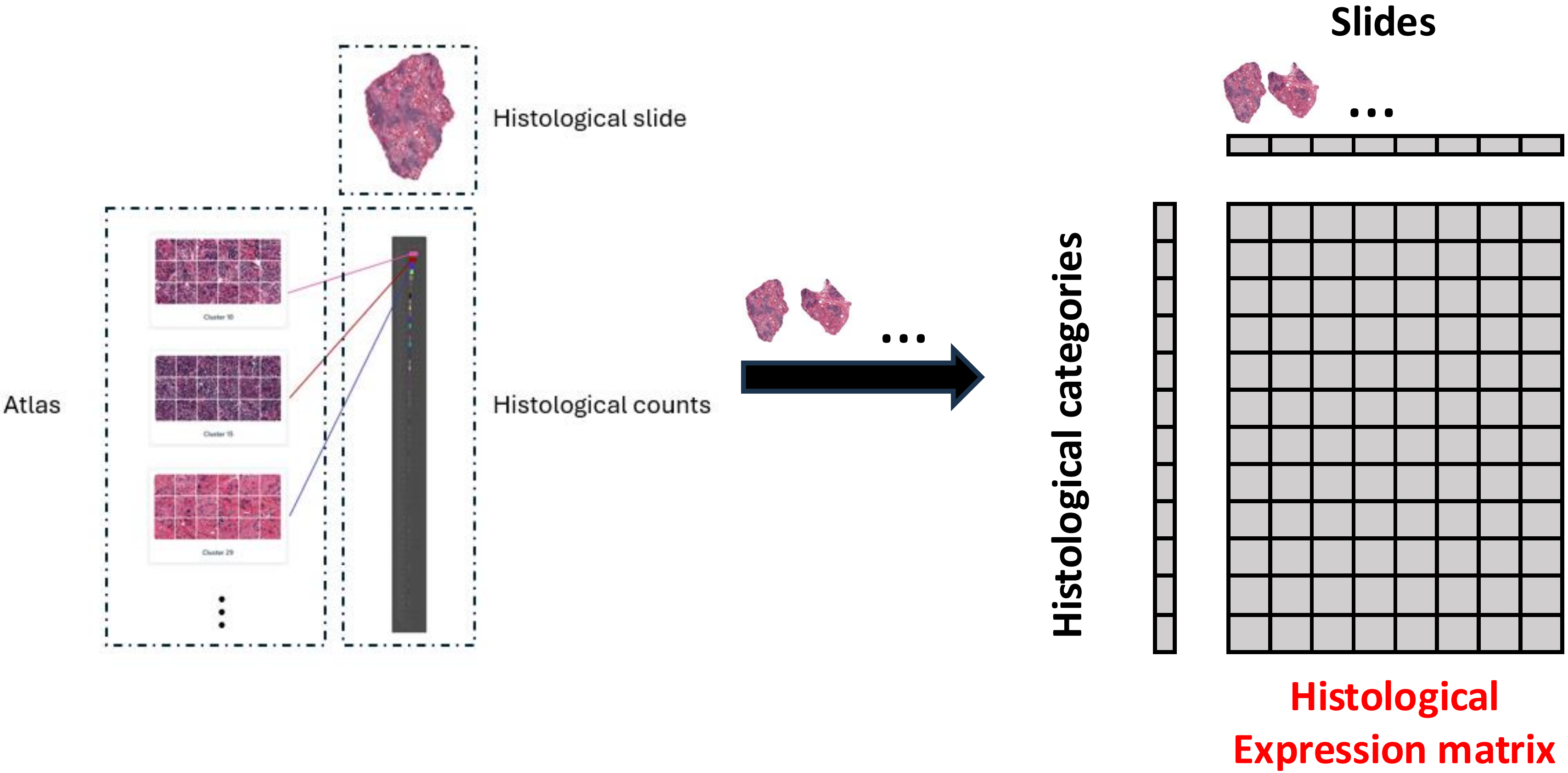
Categories projected on the slide



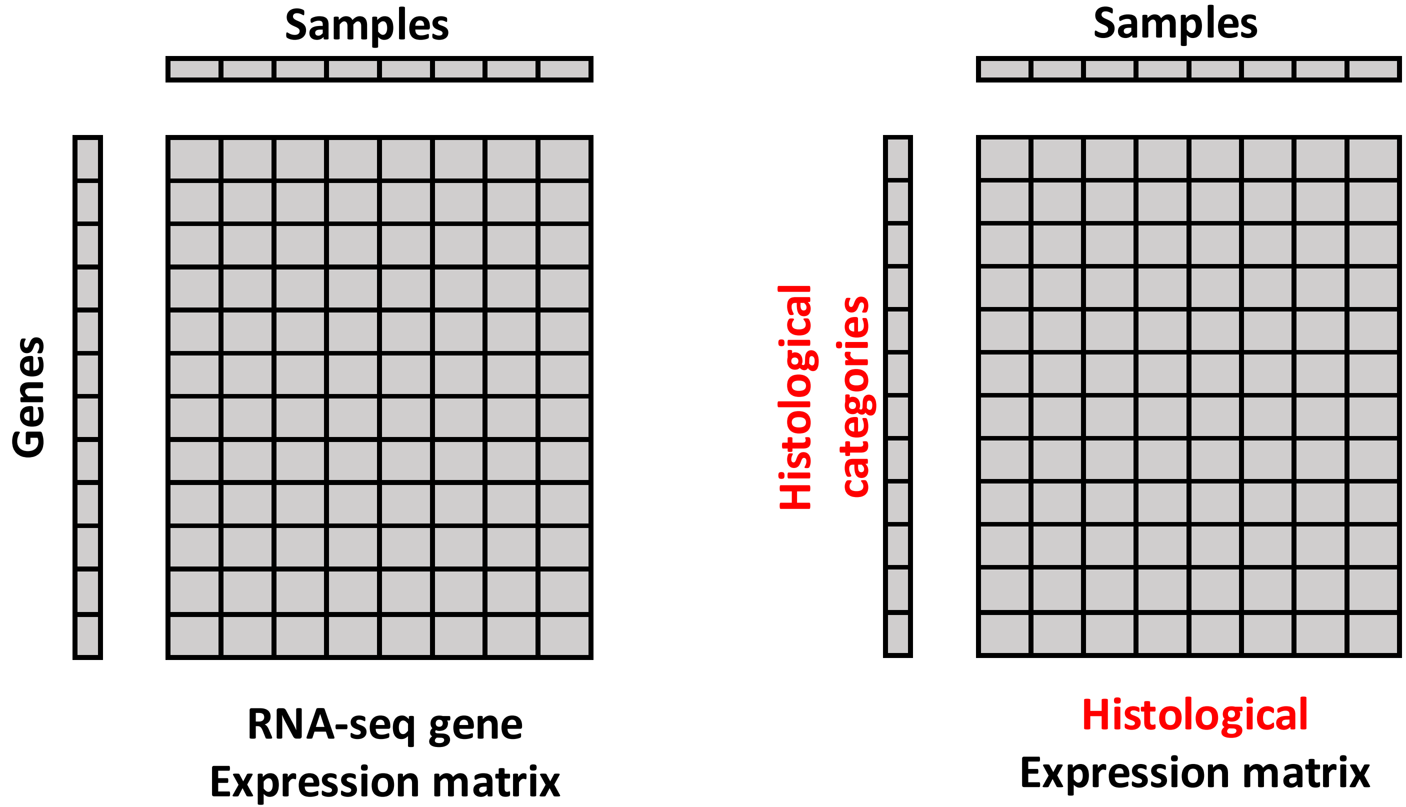
Step #4: we count tiles from each histological category in each individual slides



The histological expression matrix



Gene expression vs. histological expressio



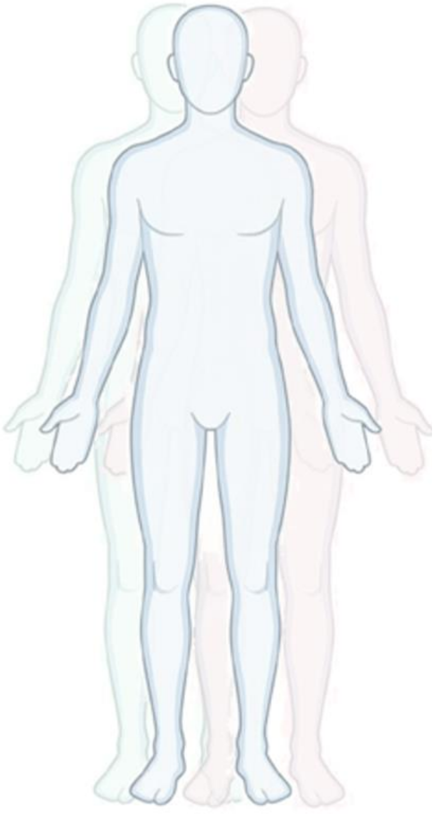
Unbiased, quantitative, histopathology

Our framework

- Is unbiased, i.e. independent of previous knowledge of histology and biomedicine
- Is rigorously quantitative and reproducible
- Brings to histopathology the computational methods developed in the last 25 yrs for gene expression analysis
- We call histological categories **morphemes**, they are the histological analogs of genes

An atlas of human histological diversity

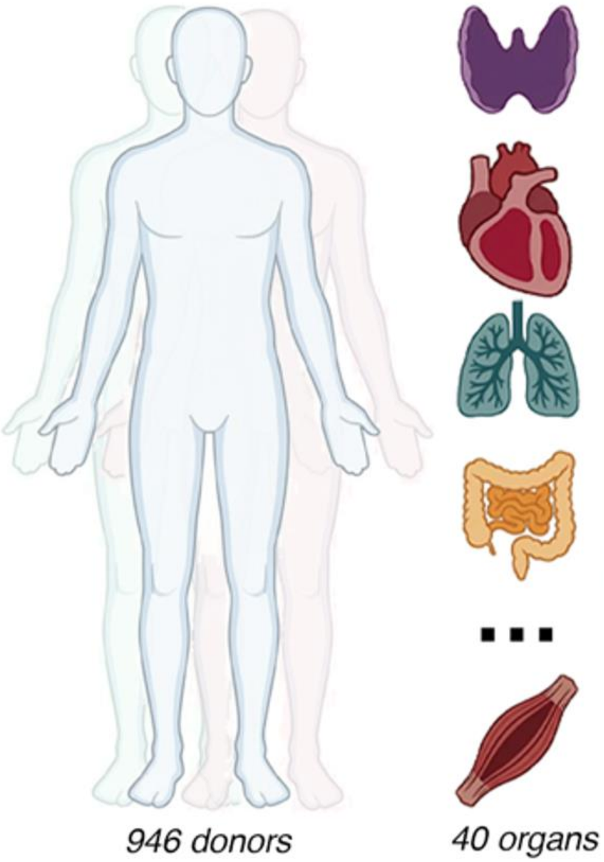
The atlas rests on the GTEx dataset



946 donors

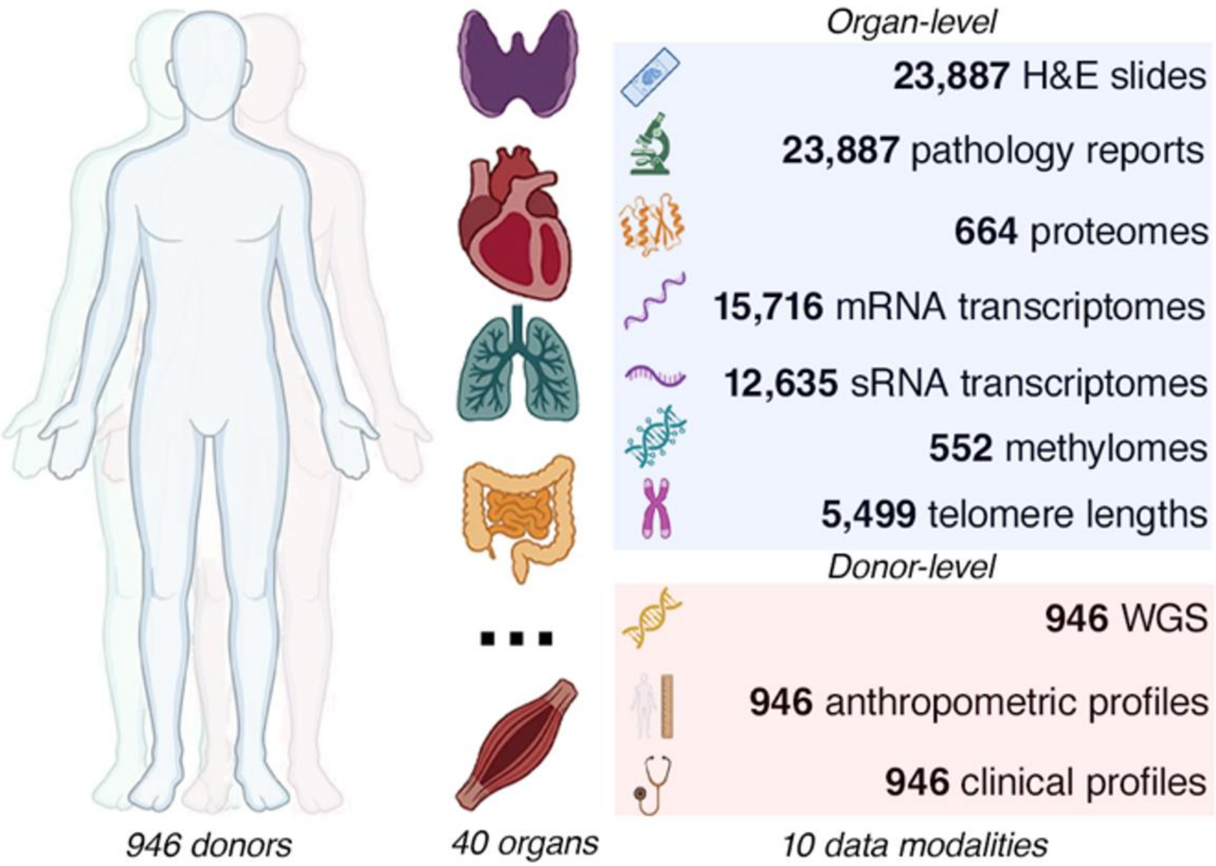
GTEx donors are ‘non-diseased’
(We will see that it’s not that simple…)

The atlas rests on the GTEx dataset



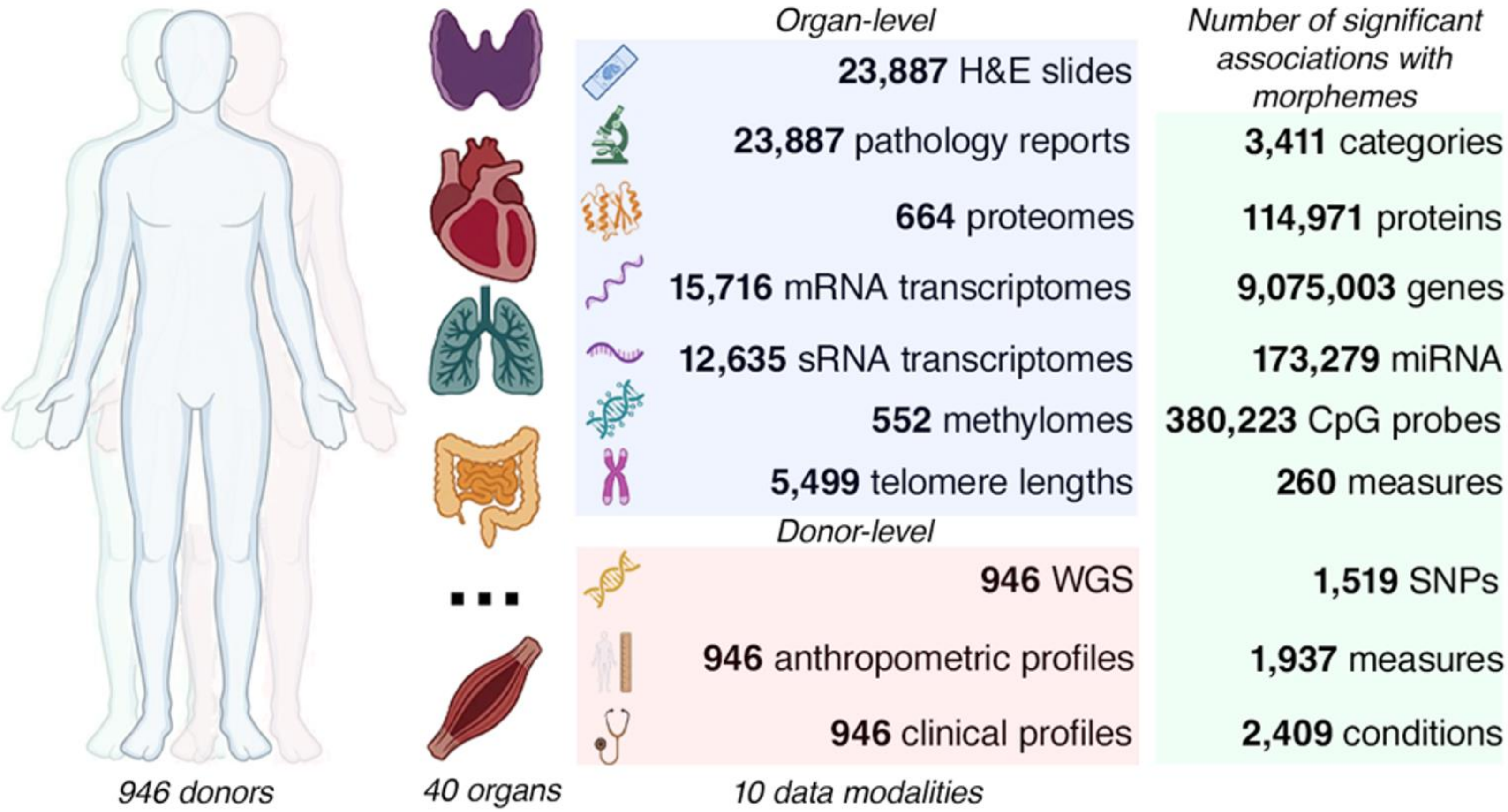
Organs were collected post-mortem

The atlas rests on the GTEx dataset



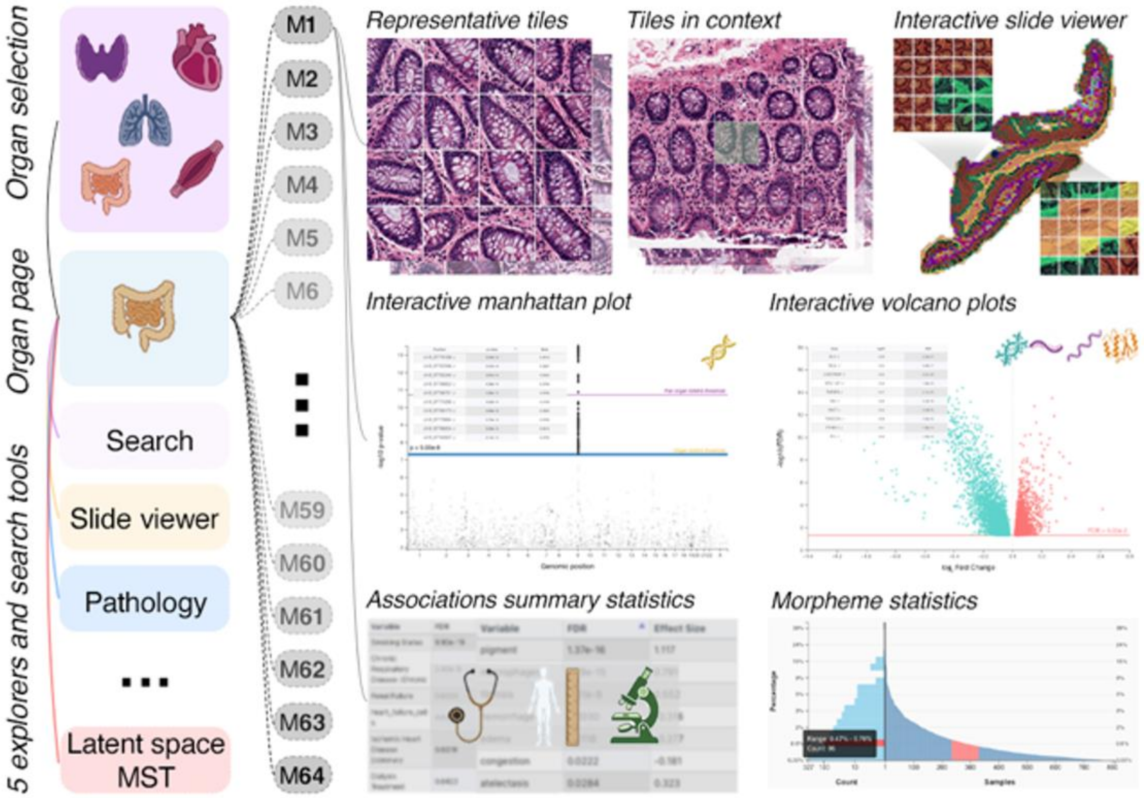
Nine data modalities are available for donors and their organs

The atlas rests on the GTEx dataset

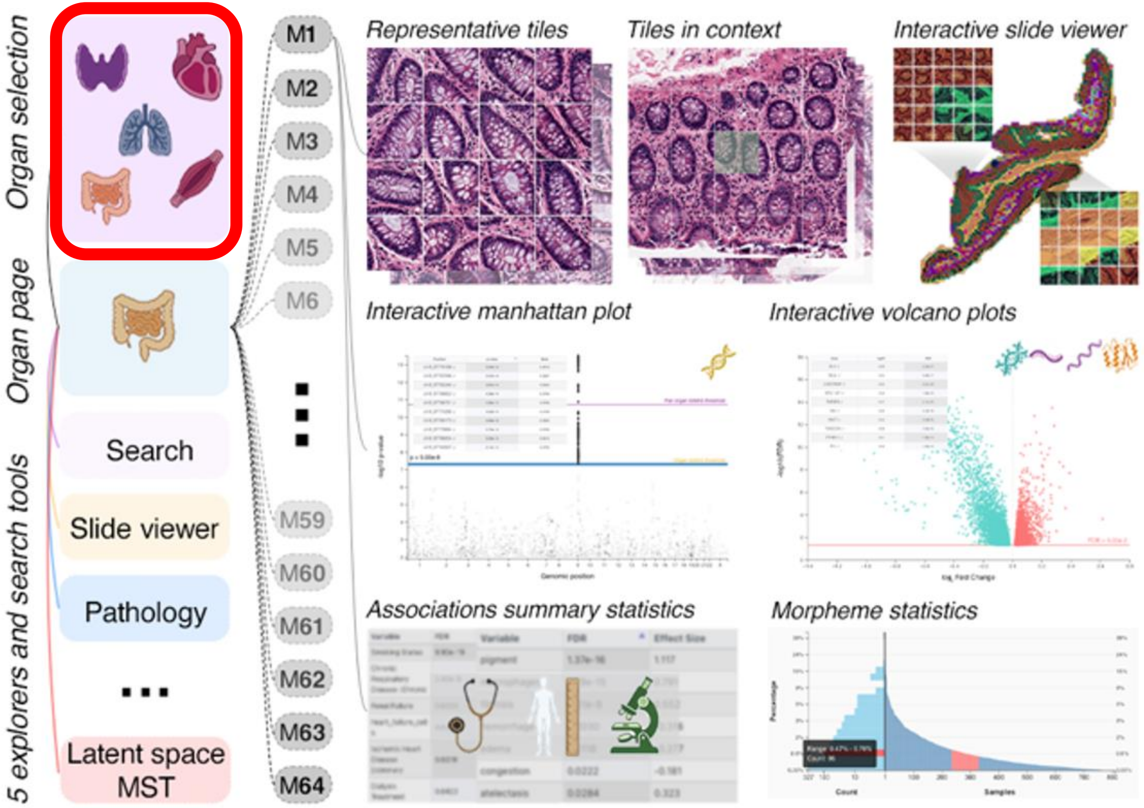


Associations with morphemes

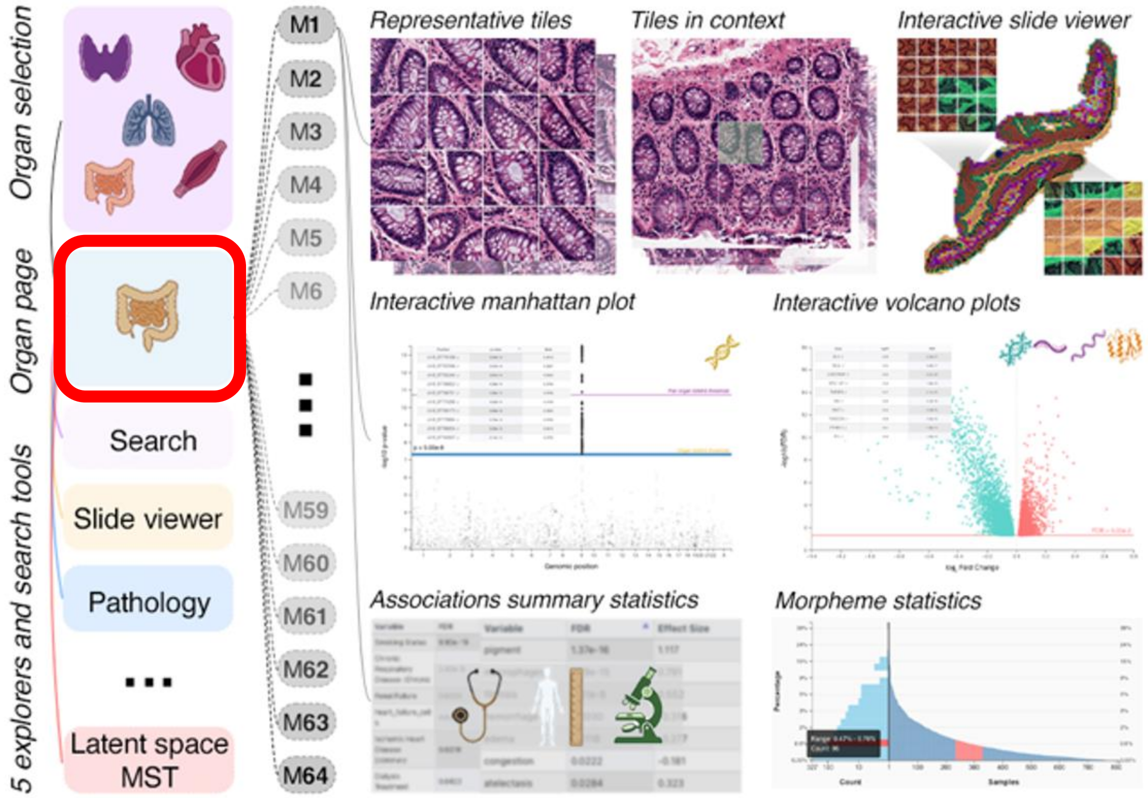
The atlas on the web



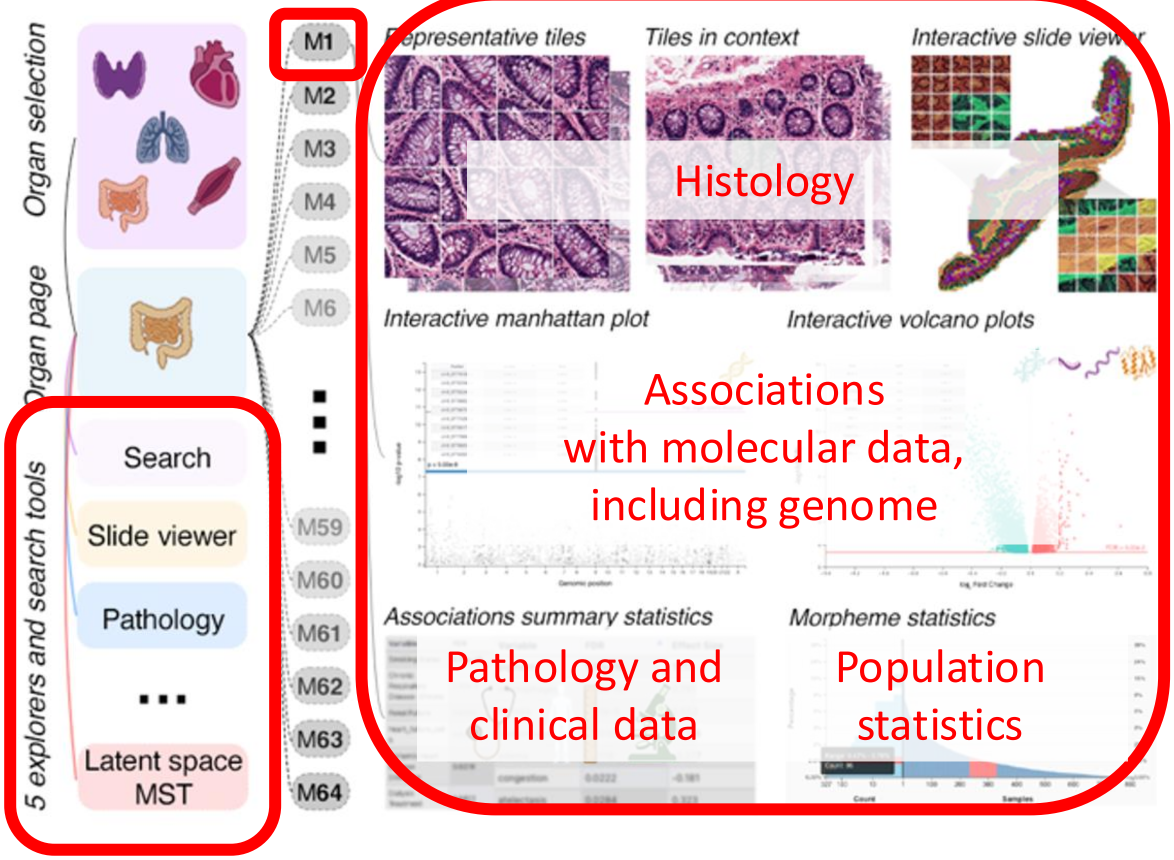
The atlas on the web



The atlas on the web



The atlas document 2,560 morphemes



The atlas is searchable with any term, including gene names

Entry points to the atlas

- Paper

https://doi.org/10.64898/2026.01.30.702249

- Online atlas

https://histologyatlas.ulb.be

- Data for your project

They will be sent to you by e-mail

Your project

Histological aging clocks

Several studies have investigated ageing effects from histological slides. The most ambitious, also based on GTEx, are arguably

- https://www.biorxiv.org/content/10.1101/2024.11.14.618081v2
- https://www.biorxiv.org/content/10.1101/2024.11.14.618081v2

Anthropometrics and technical variables

anthropometrics.csv includes,

AGE: chrological age

AGE acceleration: biological age – chronological age

SEX: 1-male, 2=female

HGHT: height of donor

WGHT: weight of donor

BMI: general indicator of the body fat an individual is carrying based upon the ratio of weight to height

Anthropometrics and technical variables

SUBJID: the GTEx ID of the subject, e.g. GTEX-111CU

COHORT: two types of donors, ‘Organ donor’ or ‘Postmortem’

TRISCHD: Ischemic Time (minutes). It’s the time elapsed between the presumed donor death and tissue collection.

DTHHRDY: Hardy scale. A number from 0 to 4 summarizing the circumstances of death

0=Ventilator Case
1=Violent and fast death
2=Fast death of natural causes
3=Intermediate death
4=Slow death

Details about annotations here:

https://ftp.ncbi.nlm.nih.gov/dbgap/studies/phs000424/phs000424.v8.p2/pheno_variable_summaries/phs000424.v8.pht002742.v8.GTEx_Subject_Phenotypes.data_dict.xml

Pathology

File pathology.csv

- GTEx pathologists examined the slides and recorded some incidental diagnostics

- These are encoded as keyworks in pathology.csv

- The number of recorded pathologies differs among organs

Telomere measurments

telomeres.csv contains telomere measurements

These are not available for all samples

Remember that they were derived from bulk samples, i.e. they are averages over billions of cells

Morpheme count matrix

- I’ll provide a matrix of morphemes count (morphome.csv)
- The columns stand for morpheme (k=64 morpheme). Ordering is the same as in the atlas. You can visualize them at https://histologyatlas.ulb.be
- The rows stand for samples (indexed by **SBJLID**)
- For example, if matrix entry (GTEX-111CU-, Morpheme-15) = 286 :
  - It means thyroid GTEX-111CU-0226 has 286 tiles in Mopheme 15.
  - Since tiles have a defined physical surface, 110x110 micron^2 (0.11x0.11 mm^2) it also mean that Mophological-cluster-23 spans 286x0.11^2=3.5 mm^2 in histological image of thyroid GTEX-111CU-0226

Gene count matrix

I'll also provide expression data (transcriptome.csv):

- There is one row per transcripts
- Transcript ENSEMBL IDs or gene symbols are in the first column
- Beside these, each column stands for a sample
- Expression was recorded as raw, unormalized, read counts, i.e. a read count of 11 for gene G means that 11 reads align on G
- Raw read counts are suitable inputs for differential gene expression software such as DESeq2 or edgeR
- Column names of the expression matrix are indexed with **SMPLID**

Your project step-by-step

Questions to be addressed

Overview of the project

In a nutshell,

The overall goal is to identify morphemes associated with age acceleration

Investigate association of age acceleration with telomere length

Investigate related gene expression programs

Q1: explore clinical variables

1. How are they distributed?

2. Are they correlated?

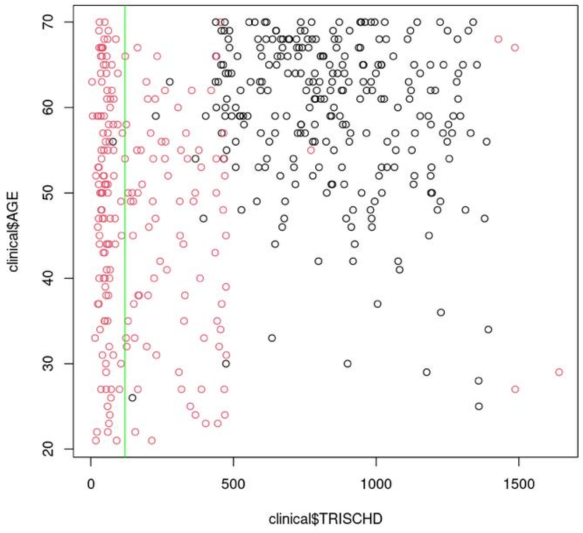
3. Do you see that some variables may confound age acceleration and telomere length?

4. If so, recompute age acceleration vs telomere length association. Discuss the result from a biological perspective

Hint: Tools to address this may include PCA analysis, showing a variable's correlation matrix, specific plots,…

Hint: **DTHHRDY** is more akin to a categorical variable.

Q1: explore clinical variables



COHORT:
Organ donors
Postmortem

Hint: Q1.3 is very important for the next steps!

An example of confounding effect: Here we see that AGE is confounded by TRISCHD and COHORT

Thus, any statistical analysis of AGE will need to be adjusted (i.e. normalized) for TRISCHD and COHORT

(Your data won't necessarily look like this…)

Q2: Associated morphemes

Taking into account your findings in Q1,

1. Compute systematically associations between age, age acceleration vs morpheme expression. Examine associations of these morphemes with anthropometrics and pathology. Are morphemes associated with age and age acceleration different?

2. Compute systematically associations between telomere length and morpheme expression. Examine associations of these morphemes with anthropometrics and pathology

3. Compare Q2.1 and Q2.2, discuss

Q2: Associated morphemes

Hint: This is a differential expression analysis where we study ‘differential **morpheme** expression’ instead of differential **gene** expression. So, I suggest you use proven tool developed for transcript count data, like DESeq2, edgeR, etc.

Hint: These tools implements multivariate model formulas that will enable the treatment of confounders in Q2.3. For example, formula `'~AGE_ACCELERATION+X+Y'` will compute the multivariate association of morphological clusters with variables AGE, X and Y. So, if you look at AGE_ACCELERATION from this model it will be adjusted for the variations of X and Y.

Thus, DESeq2 not only automatically handles normalization for total count and other intensity biases of count data, but it also enables you to cancel out your variables of choice in the analysis.

Hint: the function ‘as.formula’ (turns char string into a formula) could be useful to automate your analysis…

Q3: Association with gene expression

1. Report:
   – The number of significant down-regulated and up-regulated genes associated with age, age acceleration and telomere length
   – Report the 10 most significant down- and up-regulated genes

2. Same as Q3.1 with REACTOME gene sets (provided in the resource folder)

3. Discuss the results, technically and biologically

Q3: Association with gene expression

Hint: Use dedicated RNA-seq tools that take count data as input (e.g. DESeq2, edgeR, etc.) for Q3.1

Hint: I highly recommend you filter transcripts beforehand. You don't need transcripts that are not/little expressed in your organ. It's also a good idea to focus on transcripts showing high variability (use the median average deviation, a.k.a. mad() in R). Don't hesitate to be drastic in your filtering (e.g. it's OK to remove >50% of all transcripts)

The filtering reduces the multiple testing problem, of course. But but it also reduces the computational burden. After appropriate filtering, I could run DESeq2 analyses for all 32 morphological clusters on 465 samples in ~1 hour and a peak RAM usage < 4GB

Hint: Just as in Q2, you may need to adjust for technical confounders.

Your project step-by-step

Handing out your report

Project report

Your are asked to describe your work in a written report

It includes :

- An short introduction to the topic
- A careful description of methods and results
- A discussion of your results for each questions

Dumping code and results is *not* good enough, you must show me you understand what you are doing

Project report

- The report is handed out as a **PDF** file or a **PDF-converted Jupyter notebook**

- I am not defining a specific report length. Good reports are neither too short, nor too long. 15-20 pages (appendix not included) seems reasonable.

Project report

- This report is a good exercise for your master thesis and for scientific writing in general

- **Be specific, precise and quantitatively accurate.** For example, don’t write ‘*many genes are associated with cluster Z*’, but ‘*N genes are associated with cluster Z*’, where N is the number you have computed

Grading of your projects

- Questions Q1-Q3: **15pt**
- Quality of your PDF report (it needs to be concise, clear, precise, well presented), **3pt**
- Reproducibility: all your analyses must be reproducible, i.e. all the steps are carefully documented, including parameter settings and software versions, **2pt**
- …and **2pt** to be gained beyond 20 for the motivated ones who go off the beaten track with original analyses, etc.

Handing out your project report

- Deadline is June 7th 23:59:59.
- Send me the report by e-mail (Vincent.Detours@ulb.be), including 'BINF-F401' in the subject line
- Get back to me if I don't acknowledge receipe of your report within 3 days.

Your project step-by-step

Rules of the game

Groups

- There will be 10 groups of 2
- Each group is assigned a specific organ
- You will be informed by mail about which group you've been assigned to if you haven't chosen one by April 1st, when each group will be informed about where their data is

Rules of the game

Choose your own weapons. I do advise R, but I'll accept alternatives.

You are encouraged to communicate with one another, but

- Remember, each group has its own unique dataset
- Beware of herd effects, the majority and/or the leading figures can be dead wrong
- Plagiarism is not acceptable
- **It's up to you to use or not AI as long as you clearly explain when and how you use it.** Of course you will be held responsible for the entirety of your report.

Rules of the game

I'll address your questions by e-mail, if

- You use your brain and do your home work before asking, e.g. I won't reply if the answer is in the R documentation or can be obtained from a basic web search. (I basically request the same etiquette as in any technical forum on the Internet.)
- The subject line of all your correspondance with me must start with 'BINF-F401: '

Some tips from past experience…

- This is a science project, your technical choices must be argumented and the calculations must be reproducible, i.e. all the information needed to rerun the analyses must be provided. *In science details do matter*.
- Dumping graphics and stats in a document is not enough. The students who got the best grades examined critically their results and demonstrated that they understood the limits and biological significance of what they did.
- Technical problems are hard to anticipate, so don’t wait for the last minute to discover them, and then be left with no time to overcome them.